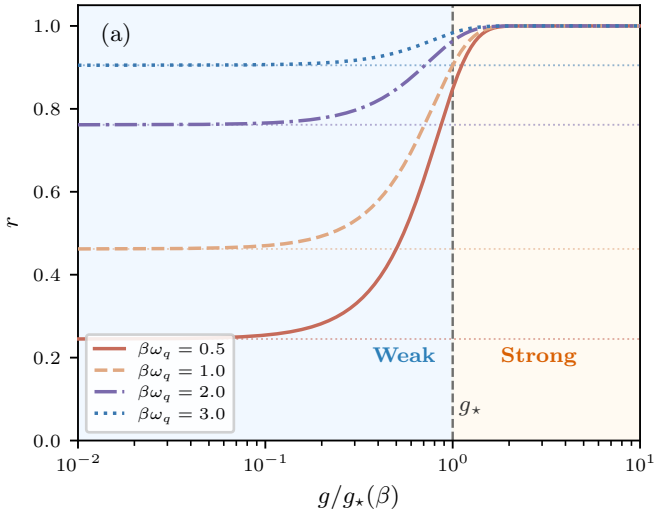
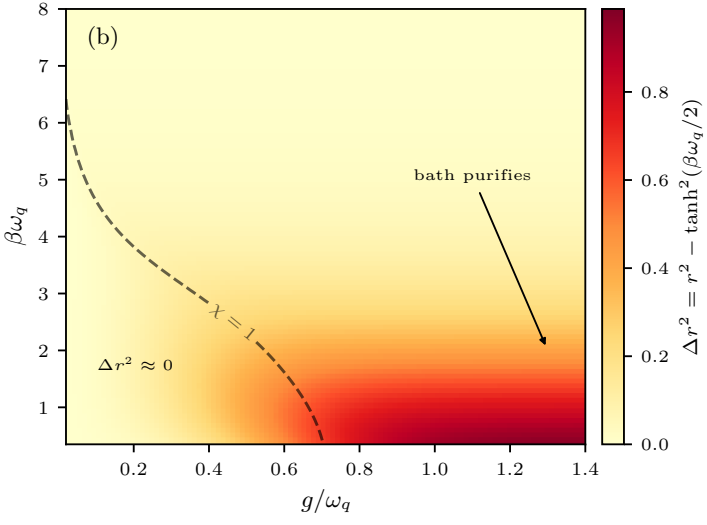

(a)
r
$g/g_\star(\beta)$
$g_\star$
Weak
Strong
$\beta\omega_q = 0.5$
$\beta\omega_q = 1.0$
$\beta\omega_q = 2.0$
$\beta\omega_q = 3.0$


(b)
$\beta\omega_q$
g/ω_q
$\chi = 1$
$\Delta r^2 \approx 0$
bath purifies
$\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$